

Why HttpServlet class is declared as abstract class BUT with 100 % concrete functionality ?

It is abstract because the implementations of key servicing methods have to be provided by (e.g. overridden by) servlet developer. Since it's abstract, its instance can't be created.

A subclass of HttpServlet must override at least one method, usually one of these:

doGet, if the servlet supports HTTP GET requests

doPost, for HTTP POST requests

doPut, for HTTP PUT requests

doDelete, for HTTP DELETE requests

init and destroy, to manage resources that are held for the life of the servlet

If you extend the class without overriding any methods, you will get a useless servlet; i.e. it will give an error response for all requests. (HTTP 405 : Method not implemented). So, if the class was not abstract, then any direct instance of HttpServlet would be useless.

So the reason for making the HttpServlet class abstract is to prevent a programming error.

As a servlet developer, you can choose to override the functionality of your requirement (eg : doPost) & ignore other methods.